Data Graph Constructions for Geodesic Distance Approximation

gflow development note

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Abstract

This report describes data-to-graph constructions for approximating intrinsic geodesic distances from finite samples. The motivating problem is to construct a weighted graph $G(X) = (V, E, \ell)$, with $V = X$, whose shortest-path metric d_G approximates geodesic distances on an underlying structured object. This requires two separate choices: an edge support that blocks false ambient shortcuts while retaining enough local connectivity, and edge lengths that represent local travel costs rather than unrelated similarities. We compare directed kNN, symmetric kNN, mutual kNN, intersection kNN, fixed-radius graphs, adaptive-radius graphs, and PHATE's thresholded adaptive affinity support. PHATE remains important here, but as an affinity and diffusion construction whose nonzero support is close to a threshold-inflated symmetric kNN or max-rule adaptive-radius graph. We also describe local geodesic pruning, older iKNN-specific global pruning, component-MST connectivity repair, and practical examples where folds, branches, near crossings, density changes, and quadratic graph surfaces stress the graph construction.

1 Introduction

Many high-dimensional data sets are not well represented by ambient Euclidean distance alone. Samples may lie near a nonlinear manifold, a curve network, a branching state space, a folded surface, a loop, a stratified object, or a noisy tube around one of these ideal geometries. In such settings, two observations can be close in $\mathbb{R}^p$ while far along the intrinsic object. This happens across folds, near self-approaches, at crossing or nearly crossing branches, and in regions where noise fills a small ambient gap. The reverse problem also appears: points that are intrinsically adjacent can be separated by sampling density variation, local curvature, or finite-sample holes.

The purpose of a data graph for geodesic reconstruction is to turn the observed sample into a finite metric proxy for intrinsic travel. A good graph allows paths to follow locally supported data geometry. A bad graph admits short circuits across unrelated nearby branches, or disconnects the sample so that finite graph distances become infinite. The central design tension is therefore local: keep enough edges to follow the object, but not so many that the graph forgets the difference between ambient proximity and intrinsic adjacency.

This report grew out of an audit of PHATE's graph construction in `gflow`. PHATE does not build an ordinary k-nearest-neighbor graph. It first constructs directed adaptive affinities, thresholds small entries, symmetrizes, and then row-normalizes the resulting kernel to obtain a diffusion operator. With the usual PHATE decay and threshold values, the nonzero undirected support is close to a slightly inflated symmetric kNN support. That observation motivates the broader comparison in this report: if the support geometry is what matters for graph shortest paths, then symmetric kNN, mutual kNN, intersection kNN, fixed-radius, and adaptive-radius graphs should be studied directly as graph-geodesic estimators.

2 The data geodesic reconstruction problem

Let

$$X = \{x_1, \dots, x_n\} \subset \mathbb{R}^p$$

be a finite sample from, or near, a structured geometric object M. When a continuous target exists, let d_M denote the intrinsic geodesic distance on M. For a smooth Riemannian manifold, for example,

$$d_M(x,y) = \inf_{\gamma(0)=x,\ \gamma(1)=y} \int_0^1 \sqrt{g_{\gamma(t)}(\dot\gamma(t), \dot\gamma(t))}\,\mathrm{d}t.$$

For branching or stratified data, d_M should be read more generally as the shortest admissible intrinsic path length on the object being sampled.

A data graph construction maps X to a weighted undirected graph

$$G(X) = (V, E, \ell), \qquad V = \{1, \dots, n\}, \qquad \ell_{ij} > 0 \text{ for } \{i,j\} \in E.$$

The graph shortest-path distance is

$$d_G(i,j) = \min_{i=v_0,\dots,v_m=j} \sum_{r=0}^{m-1} \ell_{v_r v_{r+1}},$$

with $d_G(i,j) = \infty$ when i and j lie in different connected components. The guiding reconstruction goal is

$$d_G(i,j) \approx d_M(x_i, x_j) \qquad \text{for sampled pairs } i, j.$$

In finite simulations one can also compare against a sample-oracle distance d_X^{oracle}, which asks how well the reference geodesic can be represented by paths through the observed sample points. If γ_{ij} is a reference geodesic and $(a_0, \dots, a_m)$ is an ordered set of observed sample points near it, then a typical sample-oracle target is

$$d_X^{\text{oracle}}(i,j) = \sum_{r=0}^{m-1} \|x_{a_{r+1}} - x_{a_r}\|_2.$$

The continuum target tests whether the graph recovers the underlying geometry. The sample-oracle target tests whether the graph is close to the best route that the finite observations could plausibly support.

Most graph families in the current `gflow` benchmark use ambient Euclidean edge lengths,

$$\ell_{ij} = \|x_i - x_j\|_2.$$

This is a length convention, not an affinity convention. iKNN can also carry native intersection-mediated detour lengths, and PHATE produces affinities that must not be treated as graph lengths without an explicit transformation.

3 Conventions and terminology

Let $D = (d_{ij})$ be a symmetric nonnegative input distance matrix with $d_{ii} = 0$. In Euclidean examples,

$$d_{ij} = \|x_i - x_j\|_2.$$

For $k \in \{1, \dots, n-1\}$, let $N_k(i)$ be the k nearest non-self neighbors of vertex i, using deterministic tie-breaking when needed. Define the closed neighborhood

$$\widehat{N}_k(i) = \{i\} \cup N_k(i),$$

and the local k-neighbor radius

$$\sigma_i = d_{i,(k)}.$$

When there are no nearest-neighbor ties,

$$j \in N_k(i) \quad \Longleftrightarrow \quad d_{ij} \le \sigma_i.$$

Construction	Primary object	Edge support	Default length or weight
Directed kNN	directed search graph	$i \to j \Longleftrightarrow j \in N_k(i)$	search primitive
sKNN/uKNN	undirected graph	$j \in N_k(i)$ or $i \in N_k(j)$	length $\|x_i - x_j\|_2$
mKNN	undirected graph	$j \in N_k(i)$ and $i \in N_k(j)$	length $\|x_i - x_j\|_2$
iKNN	undirected graph	$\widehat{N}_k(i) \cap \widehat{N}_k(j) \ne \varnothing$	detour or direct length
Fixed radius	undirected graph	$d_{ij} \le \varepsilon$	length $\|x_i - x_j\|_2$
Adaptive radius	undirected graph	$d_{ij} \le c\,h(\sigma_i, \sigma_j)$	length $\|x_i - x_j\|_2$
PHATE	affinity/diffusion	$K_{ij} > 0$ after thresholding	affinity, then Markov transition

4 Graph construction families

4.1 Directed kNN

The directed kNN graph is the primitive returned by a nearest-neighbor search:

$$i \to j \quad \Longleftrightarrow \quad j \in N_k(i).$$

It need not be symmetric. In graph-geodesic reconstruction, the directed lists are usually converted into an undirected support before shortest paths are computed.

4.2 Symmetric kNN

The symmetric kNN graph, also called the union kNN graph, keeps an undirected edge when at least one endpoint selects the other:

$$\{i, j\} \in E_{\mathrm{sKNN}} \quad \Longleftrightarrow \quad j \in N_k(i) \text{ or } i \in N_k(j).$$

In the no-tie case this is equivalent to the max-scale rule

$$\{i, j\} \in E_{\mathrm{sKNN}} \quad \Longleftrightarrow \quad d_{ij} \le \max(\sigma_i, \sigma_j).$$

This graph is usually better connected than mutual kNN, especially in regions where sample density changes. Its cost is that the one-sided OR rule can admit cross-density or near-fold shortcut edges.

4.3 Mutual kNN

The mutual kNN graph keeps only reciprocal neighbor relations:

$$\{i, j\} \in E_{\mathrm{mKNN}} \quad \Longleftrightarrow \quad j \in N_k(i) \text{ and } i \in N_k(j).$$

Without ties,

$$\{i, j\} \in E_{\mathrm{mKNN}} \quad \Longleftrightarrow \quad d_{ij} \le \min(\sigma_i, \sigma_j).$$

mKNN is therefore a min-scale adaptive graph. It is conservative and often useful for suppressing false local shortcuts, but it can disconnect finite samples at small k or across nonuniform density.

4.4 Intersection kNN

The intersection kNN graph uses closed neighborhoods:

$$\{i,j\} \in E_{\mathrm{iKNN}} \quad \Longleftrightarrow \quad \widehat{N}_k(i) \cap \widehat{N}_k(j) \neq \varnothing.$$

This is not equivalent to a direct radius condition on d_{ij}. Two points can be connected because both share a third nearby point, even if neither endpoint selects the other as a direct neighbor. In `gflow`, the native iKNN length for edge $\{i,j\}$ is an intersection-mediated detour,

$$\ell_{ij}^{\mathrm{iKNN}} = \min_{c \in \widehat{N}_k(i) \cap \widehat{N}_k(j)} (d_{ic} + d_{jc}).$$

If the intersection contains one of the endpoints, this can equal the direct distance d_{ij}; otherwise it records a local two-hop detour through a shared neighbor.

4.5 Fixed-radius graphs

The fixed-radius graph uses a global distance threshold:

$$\{i,j\} \in E_{\varepsilon} \quad \Longleftrightarrow \quad d_{ij} \leq \varepsilon.$$

It is the deterministic data-analysis analogue of a random geometric graph when samples are random and the same ε is applied globally. The algorithm is conceptually simple: compute candidate pair distances, keep those below ε, assign local lengths, and optionally repair connectivity. The main weakness is density sensitivity. A radius that connects sparse regions may over-connect dense or near-fold regions; a radius that blocks shortcuts in dense regions may disconnect sparse ones.

4.6 Adaptive-radius graphs

Adaptive-radius graphs replace the global radius with a local scale. Let

$$\sigma_i = d_{i,(k_{\mathrm{scale}})}.$$

For radius factor $c > 0$, `gflow` currently supports

$$\{i,j\} \in E_{\mathrm{adapt,max}} \quad \Longleftrightarrow \quad d_{ij} \leq c \, \max(\sigma_i, \sigma_j),$$

and

$$\{i,j\} \in E_{\mathrm{adapt,min}} \quad \Longleftrightarrow \quad d_{ij} \leq c \, \min(\sigma_i, \sigma_j).$$

With $c = 1$ and matching no-tie conventions, the max rule recovers sKNN support and the min rule recovers mKNN support. Values $c > 1$ inflate the adaptive radius; values $c < 1$ make the graph more conservative. This family makes explicit that sKNN and mKNN are not merely neighbor-list tricks, but special cases of local-scale radius rules.

4.7 PHATE adaptive affinity support

PHATE constructs a directed adaptive affinity rather than a length graph. With local scale $\sigma_i = b \, d_{i,(k)}$, decay $a > 0$, and threshold τ,

$$A_{ij} = \exp\left[-\left(\frac{d_{ij}}{\sigma_i} \right)^a \right], \qquad A_{ii} = 1,$$

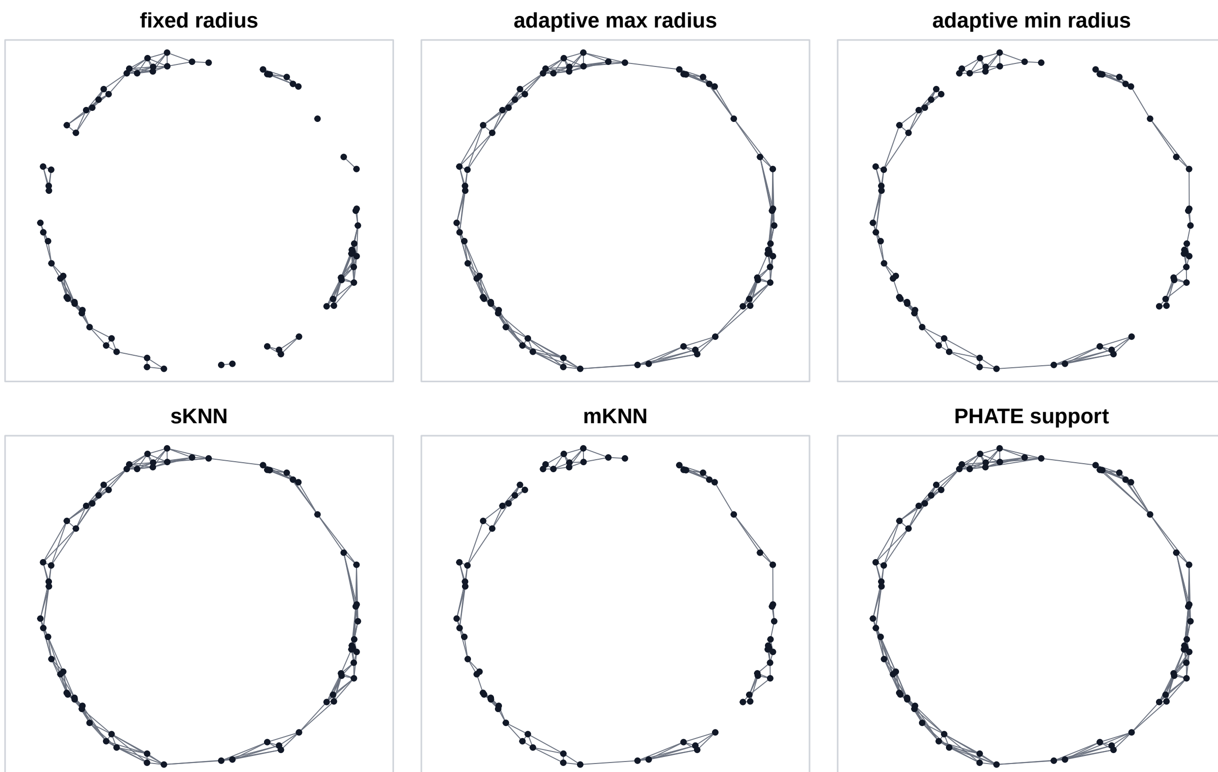

Figure 1: Graph construction rules on a nonuniform noisy circle. Fixed radius uses one global scale, adaptive max and sKNN use the larger endpoint scale, adaptive min and mKNN use the smaller endpoint scale, and PHATE support is a threshold-inflated adaptive max support.

then small directed affinities are set to zero:

$$A_{ij} = 0 \quad \text{if} \quad A_{ij} < \tau.$$

The directed survival condition is

$$d_{ij} \le r_\tau \sigma_i, \qquad r_\tau = (-\log \tau)^{1/a}.$$

For PHATE's common values $a = 40$ and $\tau = 10^{-4}$,

$$r_\tau = (-\log 10^{-4})^{1/40} \approx 1.057.$$

With additive symmetrization,

$$K_{ij} = \frac{A_{ij} + A_{ji}}{2},$$

the undirected nonzero support is

$$\{i, j\} \in E_{\mathrm{PHATE}} \quad \iff \quad d_{ij} \le r_\tau \max(\sigma_i, \sigma_j).$$

This is exactly a max-rule adaptive-radius graph support with local scales σ_i and radius factor r_τ. Thus PHATE's support is close to sKNN when r_τ is close to one. PHATE then row-normalizes K to obtain a Markov transition matrix

$$P_{ij} = \frac{K_{ij}}{\sum_{\ell=1}^{n} K_{i\ell}}.$$

The matrix P is useful for diffusion geometry and embedding, but it is not a graph-geodesic length matrix. Turning affinities into lengths requires an additional modeling choice, for example

$$\ell_{ij} = -\log(K_{ij} + \eta) \quad \text{or} \quad \ell_{ij} = K_{ij}^{-1/q},$$

and those transformations should be treated as separate research questions.

5 Support relations

Under common D, common k, no ties, PHATE additive symmetrization, and $r_\tau \ge 1$, the direct radius-like supports satisfy

$$E_{\mathrm{mKNN}} \subseteq E_{\mathrm{sKNN}} \subseteq E_{\mathrm{PHATE}}.$$

The first inclusion follows from AND implying OR. The second follows from

$$\max(\sigma_i, \sigma_j) \le r_\tau \max(\sigma_i, \sigma_j).$$

iKNN is different. It is based on shared closed neighborhoods, not on a direct radius threshold. Therefore neither

$$E_{\mathrm{iKNN}} \subseteq E_{\mathrm{PHATE}} \qquad \text{nor} \qquad E_{\mathrm{PHATE}} \subseteq E_{\mathrm{iKNN}}$$

holds in general. Empirically, iKNN can be denser than PHATE, but the relationship is not a simple nesting hierarchy.

Relationship	Holds under matching no-tie conventions?	Reason
$E_{\mathrm{mKNN}} \subseteq E_{\mathrm{sKNN}}$	yes	reciprocal implies one-sided
$E_{\mathrm{sKNN}} \subseteq E_{\mathrm{PHATE}}$	yes if $r_\tau \ge 1$	threshold inflation
$E_{\mathrm{mKNN}} \subseteq E_{\mathrm{iKNN}}$	yes for closed iKNN	endpoints are shared
$E_{\mathrm{iKNN}} \subseteq E_{\mathrm{PHATE}}$	no	shared-neighbor predicate
$E_{\mathrm{PHATE}} \subseteq E_{\mathrm{iKNN}}$	no	direct adaptive radius predicate

6 Pruning graph supports

Graph construction often produces edges that are locally redundant or geometrically suspicious. Pruning is a post-construction sparsification stage. It should not be conflated with the original support rule: a pruned sKNN graph and a native mKNN graph can have similar edge counts while representing different local decisions.

6.1 Local geodesic pruning

Local geodesic pruning is the main cross-family pruning method currently used in `gflow`. Let $G = (V, E, \ell)$ be a candidate graph. For an edge $\{u, v\}$ with length ℓ_{uv}, define a local vertex set $L(u, v)$. In the current implementation,

$$L(u, v) = \{u, v\} \cup N_{k_{\text{local}}}(u) \cup N_{k_{\text{local}}}(v),$$

where the local neighborhoods are computed from the ambient sample coordinates. Temporarily exclude the edge $\{u, v\}$ and compute the shortest alternative path from u to v inside the induced local graph:

$$d_{L(u,v)\setminus\{uv\}}(u, v) = \min_{\pi: u \rightsquigarrow v,\ \pi \subseteq L(u,v),\ \{u,v\} \notin \pi} \sum_{\{a,b\} \in \pi} \ell_{ab}.$$

For tolerance $\tau_{\text{prune}} > 1$, prune the edge when

$$d_{L(u,v)\setminus\{uv\}}(u, v) \leq \tau_{\text{prune}}\, \ell_{uv}.$$

Edges are processed longest-first. Intuitively, a direct edge is removed when nearby retained edges already provide a path that is only slightly longer. This is spanner-like in spirit, but it is a pragmatic local data-graph rule rather than a proof-carrying global spanner construction.

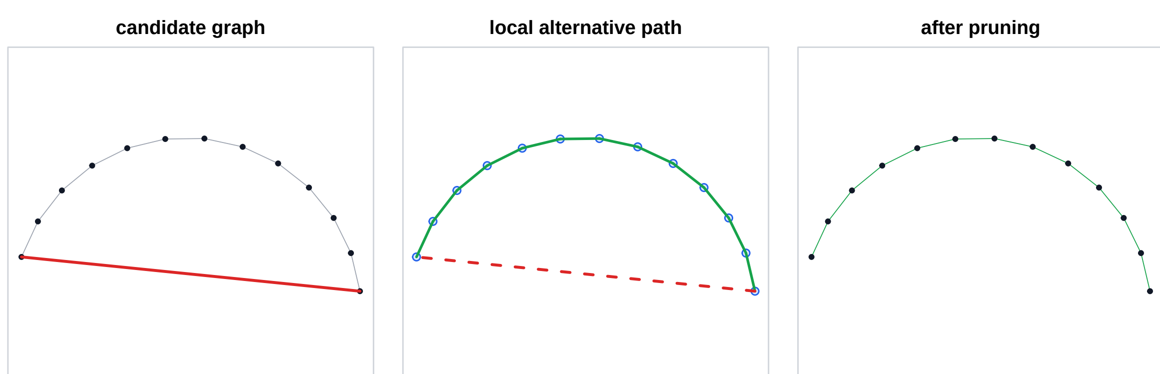

Figure 2: Local geodesic pruning. A long candidate chord is tested against an alternative path restricted to a local neighborhood. If the alternative path length is at most $\tau_{\text{prune}}\ell_{uv}$, the chord is locally redundant and can be removed.

6.2 iKNN global geometric pruning and long-edge pruning

`create.single.iknn.graph()` also retains older iKNN-specific global geodesic pruning. It considers candidate long edges and removes an edge when the whole current graph contains an alternative path whose path-to-edge ratio is below a threshold. In notation, one tests whether

$$\frac{d_{G\setminus\{uv\}}(u,v)}{\ell_{uv}} \le 1+\delta.$$

This is not the same algorithm as local geodesic pruning: the alternative path is global, candidate selection can be percentile-based, and it historically belongs to the iKNN path.

There is also quantile-style long-edge pruning in the shared pruning helper. It targets edges above an edge-length percentile and removes a long edge only if its removal does not disconnect the currently retained graph. This is best understood as a conservative long-edge cleanup step, not as a substitute for local geometric pruning.

6.3 Pruning order and lifecycle stages

For geodesic reconstruction experiments, disconnected graphs create infinite distances. `gflow` therefore records multiple lifecycle stages so benchmarks can compare construction, pruning, and repair order:

$$\text{raw},\quad \text{raw.repaired},\quad \text{pruned},\quad \text{pruned.repaired},\quad \text{repaired.pruned},\quad \text{final}.$$

The current quadratic-surface benchmark favors

$$\texttt{raw.repaired}\quad \text{when pruning is disabled,}$$

and

$$\texttt{repaired.pruned}\quad \text{when local pruning is enabled.}$$

This compares finite graph-geodesic matrices while still asking whether local pruning can remove redundant or shortcut-like edges after connectivity has been stabilized.

7 MST connectivity repair

Neighborhood graphs can be disconnected. Sometimes this is meaningful: separate components may reflect genuinely separate objects. But graph geodesic distances, diffusion operators, spectral methods, and Isomap-style procedures often require a connected graph. Connectivity repair should therefore be explicit and diagnosable.

Let an initial graph have connected components

$$C_1,\dots,C_m.$$

For each component pair, define the shortest possible Euclidean bridge

$$w(C_a,C_b)=\min_{i\in C_a,\ j\in C_b}\|x_i-x_j\|_2,$$

and choose an attaining data edge

$$(i^*_{ab},j^*_{ab})\in \operatorname*{arg\,min}_{i\in C_a,\ j\in C_b}\|x_i-x_j\|_2.$$

Construct the complete component graph with vertices $C_1,\dots,C_m$ and edge weights $w(C_a,C_b)$. A component-MST repair computes a minimum spanning tree T_C on this component graph and adds the selected bridge edges

$$\{i^*_{ab},j^*_{ab}\},\qquad (C_a,C_b)\in T_C.$$

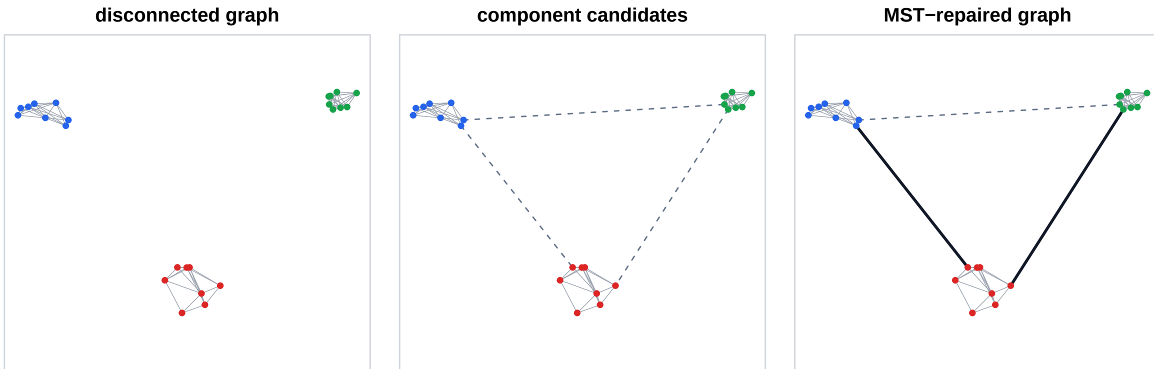

Figure 3: Component-MST repair. Components are first identified, candidate bridge lengths are computed between component pairs, and only the component-MST bridge edges are added to connect the graph.

When there are m components, this adds exactly $m-1$ bridge edges. It does not add every shortest edge between every component pair; it connects the component graph minimally.

`gflow` exposes three repair modes. `component.mst` computes exact shortest intercomponent bridges. `component.mst.ann` first tries sparse approximate nearest-neighbor bridge candidates and falls back to exact repair when needed. `global.mst` unions the graph with the full Euclidean MST on all vertices. The component-MST method is the most direct choice when the goal is to add the minimum number of diagnostic bridge edges.

8 Examples and failure modes

8.1 Finite circle examples

The root-of-unity examples show clean finite support relationships. Points are

$$x_m = (\cos(2\pi m/n), \sin(2\pi m/n)), \qquad m = 0, \ldots, n-1.$$

For $n = 3, 4, 5$, $k = 2$. For $n = 100$, $k = 10$. All constructions use Euclidean distances.

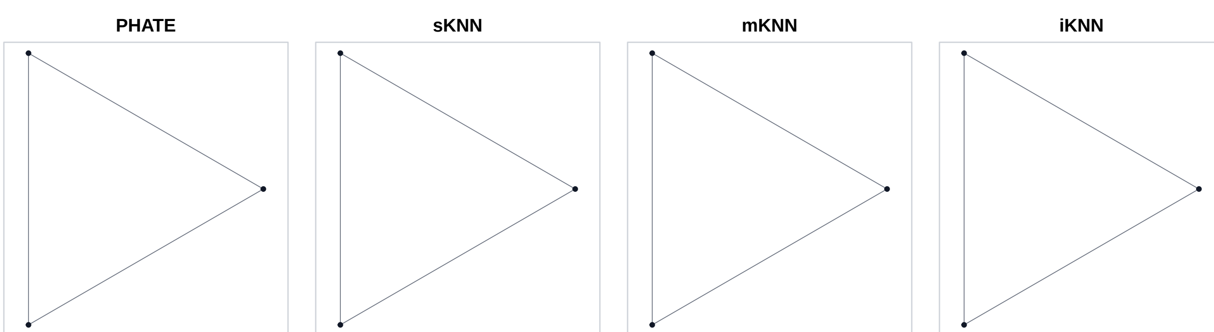

Figure 4: Three roots of unity with $k = 2$. All four constructions produce the complete graph.

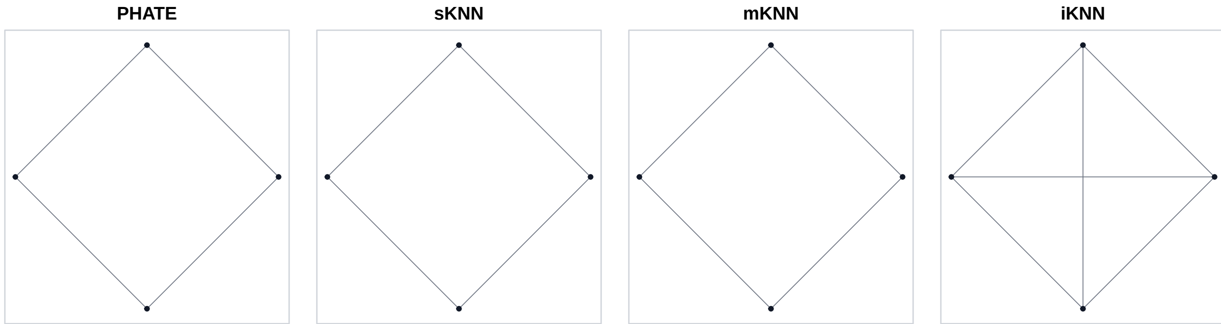

Figure 5: Four roots of unity with $k = 2$. PHATE, sKNN, and mKNN produce the cycle, while closed-neighborhood iKNN produces the complete graph.

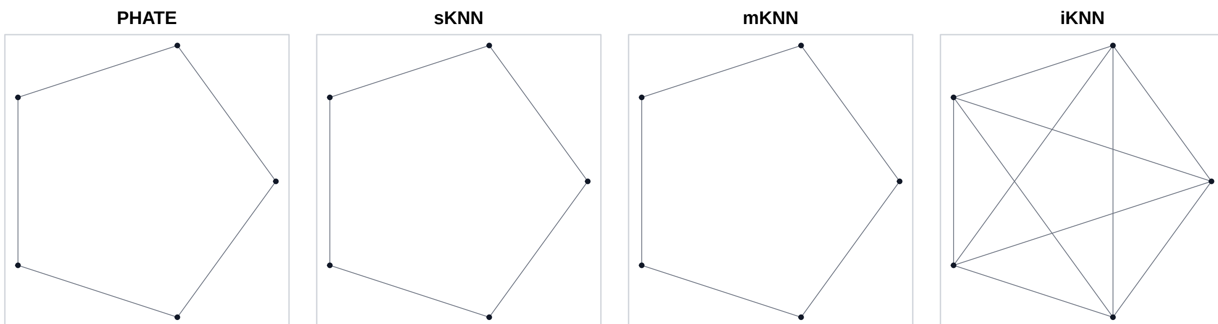

Figure 6: Five roots of unity with $k = 2$. PHATE, sKNN, and mKNN keep adjacent circle edges; closed-neighborhood iKNN connects every pair.

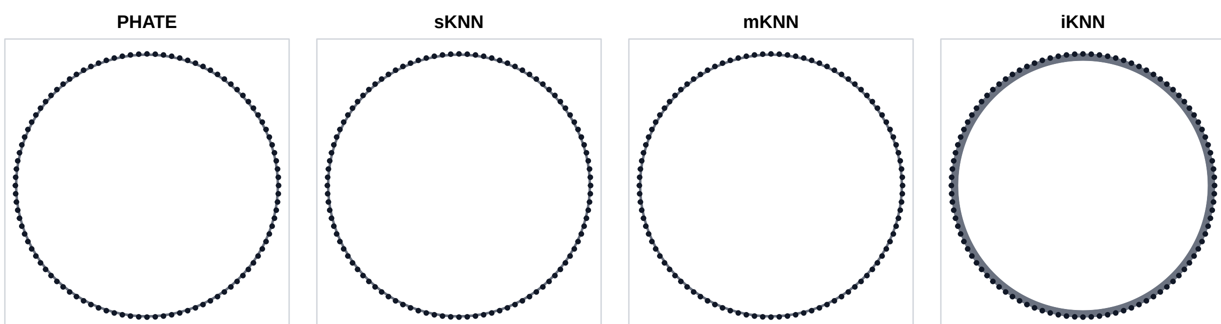

Figure 7: One hundred roots of unity with $k = 10$. PHATE remains close to sKNN, mKNN is reciprocal, and iKNN is denser because shared neighborhoods reach farther than direct adaptive-radius tests.

8.2 Noisy nonuniform circle and sample Fermat distances

The noisy-circle examples use a sample Fermat distance as the input D. Starting from a base kNN graph in observed Euclidean coordinates, each base edge has cost

$$c_{ij} = \|x_i - x_j\|_2^{\alpha}, \qquad \alpha = 2,$$

and the sample Fermat distance is

$$d_F(i,j) = \min_{\gamma: i \rightsquigarrow j} \sum_{\{u,v\} \in \gamma} c_{uv}.$$

This powered shortest-path distance penalizes long jumps and can reduce the influence of noisy ambient shortcuts before the second-stage graph support is constructed.

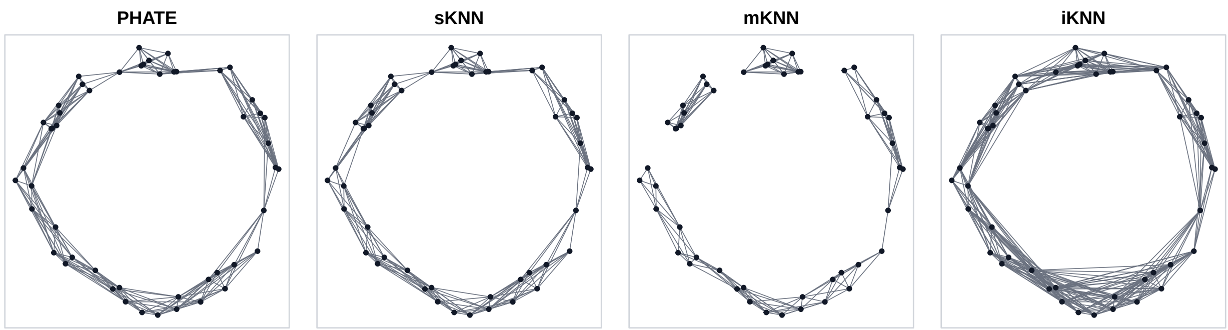

Figure 8: Noisy circle, $n = 50$, graph $k = 8$, sample Fermat distance with $\alpha = 2$.

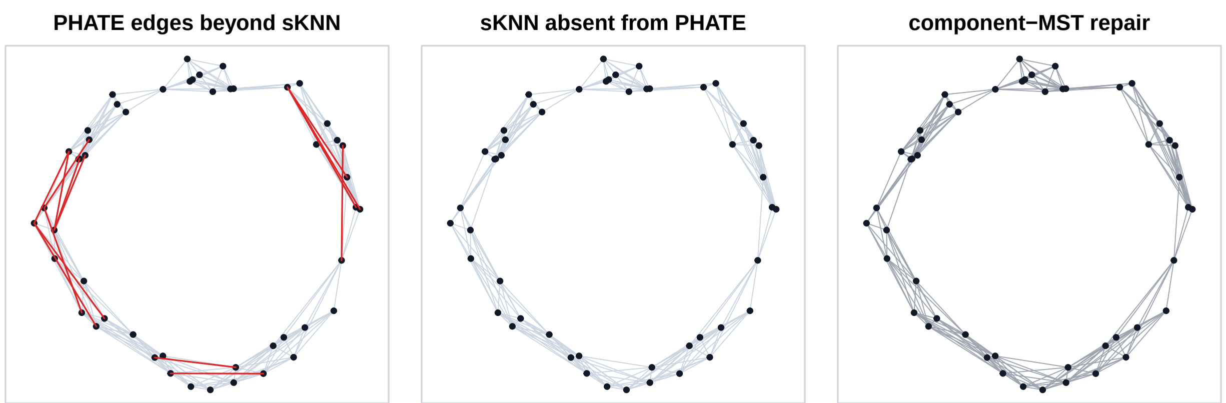

Figure 9: Noisy circle, $n = 50$, PHATE-versus-sKNN diagnostics and component-MST bridge view.

8.3 Additional 2D stress examples

Several simple 2D data sets expose distinct failure modes. A figure eight stresses self-approach. Crossing and nearly crossing line segments separate ambient proximity from branch membership. A branching tree tests whether a graph preserves arms without creating shortcuts. A nonuniform circle shows why global radius rules are density-sensitive. A quadratic parameter disk connects the report to the current surface benchmark.

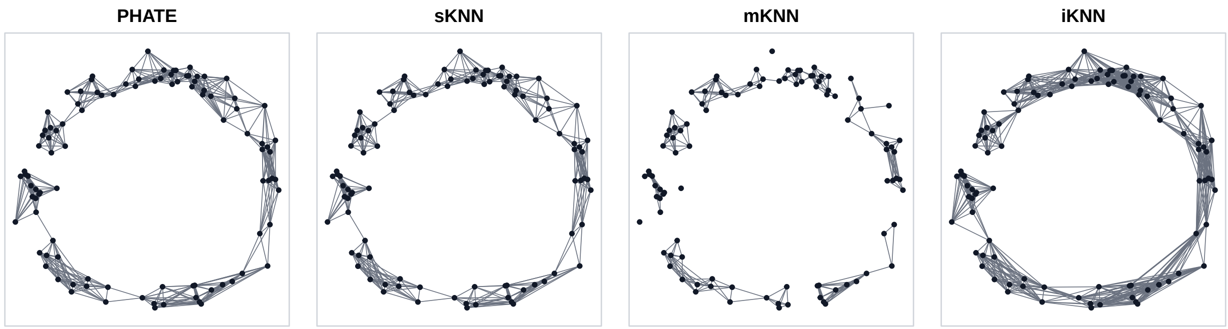

Figure 10: Noisy circle, $n = 100$, graph $k = 8$, sample Fermat distance with $\alpha = 2$.

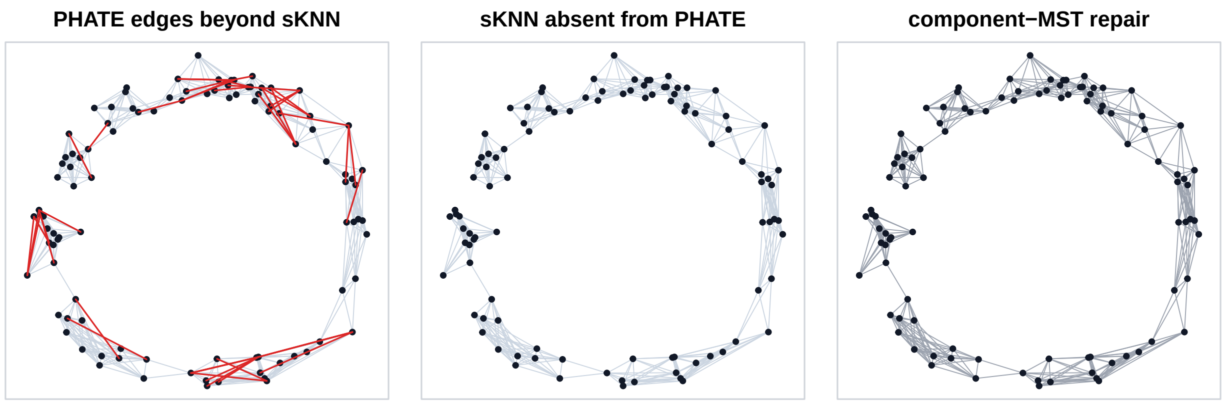

Figure 11: Noisy circle, $n = 100$, PHATE-versus-sKNN diagnostics and component-MST bridge view.

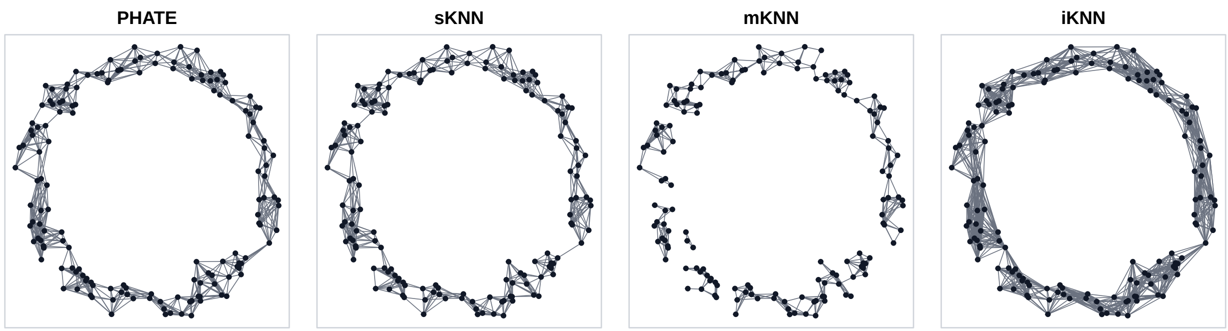

Figure 12: Noisy circle, $n = 150$, graph $k = 8$, sample Fermat distance with $\alpha = 2$.

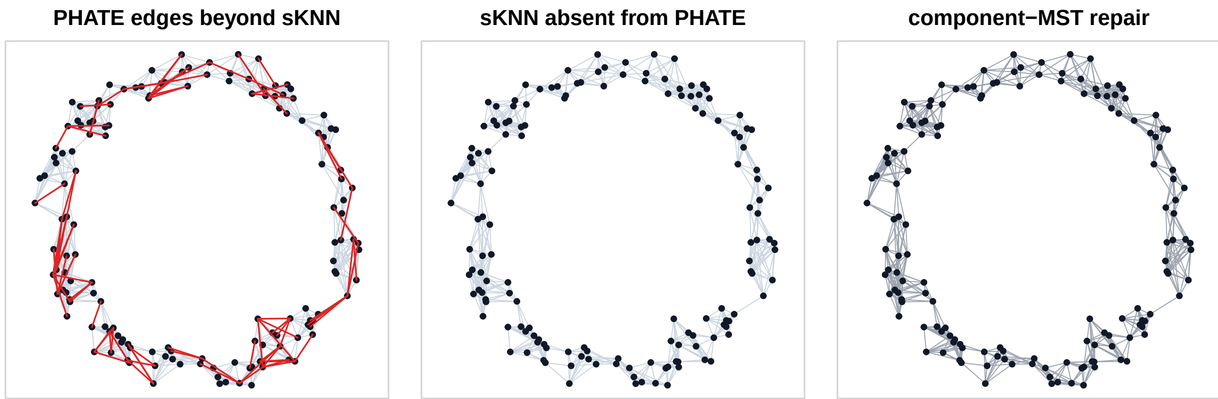

Figure 13: Noisy circle, $n = 150$, PHATE-versus-sKNN diagnostics and component-MST bridge view.

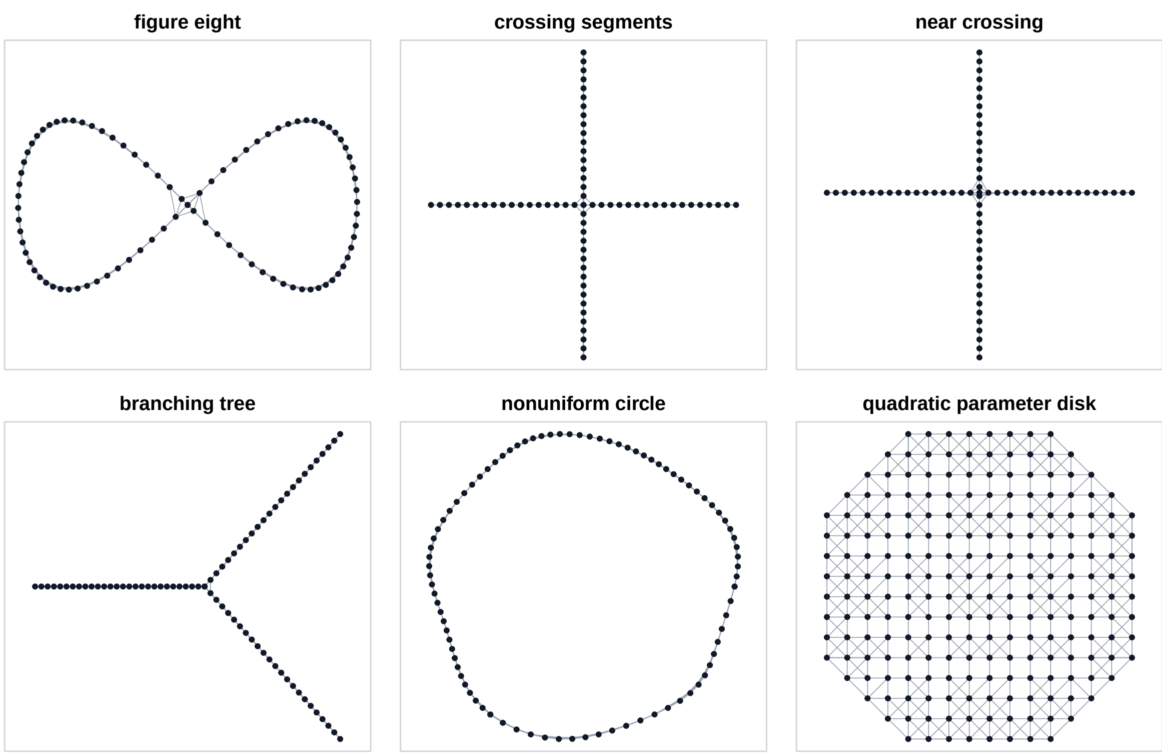

Figure 14: Illustrative 2D stress cases for data graph construction. The displayed edges are sKNN supports and are meant as visual examples of possible shortcut and connectivity issues, not benchmark conclusions.

8.4 Quadratic graph surfaces

The current geodesic benchmark focuses on two quadratic graph surfaces over the parameter disk

$$\Omega = \{(u, v) \in \mathbb{R}^2 : u^2 + v^2 \leq 1\},$$

embedded as

$$F(u, v) = (u, v, q(u, v)).$$

The two target surfaces are

$$q_{\text{paraboloid}}(u, v) = u^2 + v^2, \qquad q_{\text{saddle}}(u, v) = u^2 - v^2.$$

They are simple enough to support numerical reference geodesics while still curved enough to reveal distortion in graph shortest-path distances.

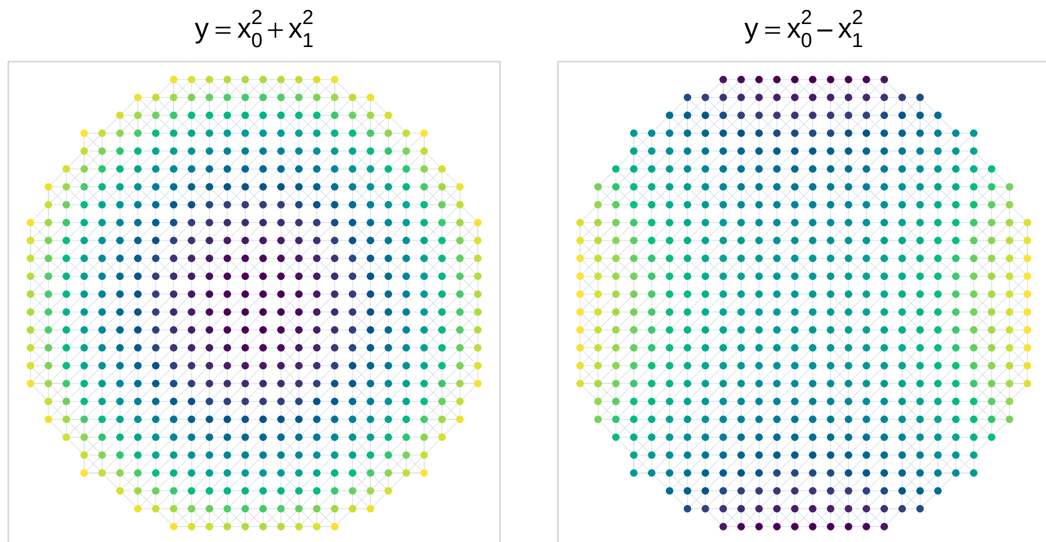

Figure 15: Quadratic graph-surface benchmark examples shown in parameter coordinates, with color indicating the graph height $q(u, v)$. The benchmark constructs data graphs on embedded points $F(u, v)$ and compares graph shortest paths to reference surface geodesics.

9 Evaluation metrics

Because graph distances can differ from reference distances by a global scale, the benchmark uses a scale-adjusted comparison. Given finite distances $d_G(i, j)$ and a reference $d_{\text{ref}}(i, j)$, define

$$\alpha^* = \arg\min_{\alpha>0} \sum_{i<j} \left(\alpha\, d_G(i, j) - d_{\text{ref}}(i, j)\right)^2.$$

The relative RMS error is

$$\left(\frac{\sum_{i<j} (\alpha^* d_G(i, j) - d_{\text{ref}}(i, j))^2}{\sum_{i<j} d_{\text{ref}}(i, j)^2} \right)^{1/2}.$$

Additional diagnostics include relative absolute error, distortion quantiles, distance correlations, component counts, edge counts, bridge counts, and pruning statistics. These metrics should always report which graph lifecycle stage and which reference target were used.

10 Computational considerations

Exact pairwise construction is deterministic and simple but costs $O(n^2)$ memory or time for the candidate distances. kNN and adaptive-radius graphs can use nearest-neighbor search to reduce the candidate set, although exact tie behavior may differ between exact and approximate search backends. Fixed-radius graphs are especially sensitive to the cost of discovering all pairs within ε at large n.

Component-MST repair has two costs: finding connected components and finding shortest inter-component bridges. Exact component bridges can be expensive if there are many large components, while approximate-neighbor bridge candidates are faster but require fallback logic when they do not connect the component graph. Local geodesic pruning is currently the most expensive benchmark stage because it repeatedly computes alternative local shortest paths while modifying the graph. That performance issue should be profiled separately before large pruning sweeps are treated as routine.

11 Historical perspective

Isomap made graph shortest paths on neighborhood graphs central to nonlinear dimensionality reduction. Its basic estimator builds a kNN or fixed-radius graph, weights local edges by ambient distance, and uses graph shortest paths as approximate manifold geodesics before classical multi-dimensional scaling. Theoretical work around Isomap and random geometric graphs clarified the asymptotic promise and the central scale tradeoff: too small a neighborhood disconnects the graph, too large a neighborhood creates short circuits.

Fixed-radius graphs connect this data-analysis practice to random geometric graphs and Gilbert's random plane network model. kNN, symmetric kNN, and mutual kNN became standard choices in clustering, semi-supervised learning, manifold learning, and spectral methods. Mutual kNN is often used as a conservative alternative to union kNN when one wants reciprocal local support.

Adaptive local scaling entered spectral clustering as a way to avoid one global bandwidth. The same idea appears here as adaptive-radius support: use $d_{i,(k)}$ as a local scale and compare d_{ij} to a function of endpoint scales. PHATE uses a related adaptive kernel but then follows an affinity-to-diffusion path rather than a graph-length path. Shared-nearest-neighbor graphs are another related but distinct lineage: they weight edges by overlap in neighbor lists, as in the Jarvis-Patrick idea and modern single-cell SNN workflows. `gflow`'s iKNN support also uses neighborhood intersections, but its edge predicate and native detour length should not be confused with Jaccard-style SNN weights.

Pruning belongs to a broader geometric sparsification tradition. Graph spanners ask for sparser subgraphs that approximately preserve distances. Local geodesic pruning borrows that intuition, but in this report it remains an empirical data-graph procedure: remove locally replaceable edges and then measure whether geodesic reconstruction improves. MST repair has a different role. It is a connectivity scaffold, grounded in classical minimum spanning tree algorithms, and should be reported as an explicit intervention rather than a native neighborhood relation.

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